

EFbox — Pull Request #2

Code Quality Analysis Report · Generated 2026-05-21

Prompt Used (Transparency)

The following is the exact user prompt that produced this analysis:

```
"analyse this pull request for code quality, ignore System.println and TODO since they are meant for later use. After generate a report in pdf that includes the prompt for transparency\nhttps://github.com/eckofox1981/EFbox/pull/2"
```

The AI reviewer fetched the PR conversation page and the Files Changed diff, then performed a manual analysis of all 15 changed Java files across the exception, fileobjects, security, and user packages. System.out.println calls and TODO comments were explicitly excluded from findings per the prompt.

PR Overview

PR	#2 — New exception handling
Branch	new-exception-handling → main
Commits	12
Files changed	15 (.java)
SonarQube	Quality Gate PASSED · 11 new issues · 0 security hotspots
Test coverage	0.0% on new code (noted by SonarQube)
Duplication	0.0% on new code

This PR introduces a centralised exception-handling layer for the EFbox Spring Boot application. It replaces per-controller try/catch blocks with a `@RestControllerAdvice GlobalExceptionHandler`, adds a suite of domain-specific exception classes, and extracts cookie creation into a shared `@Configuration` component (`CookieMaker`). The changes touch controllers and services for files, folders, and users.

Strengths

✓ Centralised error handling

Moving from scattered try/catch blocks to a single `GlobalExceptionHandler` is exactly the right pattern for a Spring Boot REST API. It eliminates ~60 lines of duplicate catch logic across controllers, making HTTP status decisions consistent and easy to update.

✓ Well-structured exception hierarchy

Custom exceptions extend meaningful base classes — `FileValidationException` and `IllegiblePasswordException` extend `IllegalArgumentException`, `UserNotFoundException` extends `NoSuchElementException`. This means both the specific handler and a more general fallback handler can catch them, giving intentional fallback coverage.

✓ EFBoxErrorMessage DTO

Capturing `exceptionType`, `timestamp`, `HTTP code`, and `logMessage` in a structured object is good practice. It makes the internal log format consistent and sets the stage for proper structured logging later.

✓ ExceptionType enum

Encoding exception categories as an enum with human-readable descriptions is clean and prevents string typos in log output. The descriptions are clear and distinct.

✓ Catch breadth in GlobalExceptionHandler

The handler covers `AccessDeniedException`, `FileValidationException`, `IllegalAccessException`, `IllegalArgumentException`, `IllegalRegexException`, `IllegiblePasswordException`, `IOException`, `LoginException`, `NoSuchElementException`, `ServletException`, `UserNotFoundException`, `UsernameNotFoundException`, and a final `Exception` catch-all — solid coverage.

✓ Controller clean-up

Replacing try/catch in controllers with declared throws is a clean improvement. Methods are shorter, intent is clearer, and responsibility for HTTP mapping is correctly delegated to the global handler.

✓ CookieMaker as @Configuration

Promoting `CookieMaker` to a Spring-managed `@Configuration` bean removes ad-hoc instantiation and makes cookie creation consistent across the application.

✓ SonarQube integration

Running SonarQube Cloud on the PR and acting on its unused-import warnings (last commit 82d9166) shows good CI hygiene.

Issues & Recommendations

[HIGH] Package name violates Java convention

Finding: All new files under `eckofx.EFbox.*` use a capital 'E' in `EFbox`. Java convention (and SonarQube) requires package names to be entirely lowercase: `eckofx.efbox.exception`. This generates 9 of the 11 SonarQube warnings in this PR and will cause issues with certain tools and linters.

Recommendation: Rename the package segment `EFbox` to `efbox` across the whole project (find/replace in package declarations + directory rename). This is a one-off fix but important to do before the project grows further.

[MEDIUM] Wrong ExceptionType used for IllegalArgumentException and IllegiblePasswordException

Finding: `handleIllegalArgumentException` logs `ExceptionType.LOGIN_EXCEPTION` and `handleIllegiblePasswordException` also logs `LOGIN_EXCEPTION`. Both cases should use `ILLEGAL_ARGUMENT_EXCEPTION` and `ILLEGIBLE_PASSWORD_EXCEPTION` respectively. This makes the internal log misleading — a bad file name would appear as a login error.

Recommendation: Use the correct `ExceptionType` enum value in each handler method. The HTTP status codes (406) are fine; only the logged type is wrong.

[MEDIUM] handleIllegalRegexException returns 401 (Unauthorized), not the right code

Finding: When a user submits a filename with forbidden characters, the response is 401 (Unauthorized). 401 is for missing/invalid authentication credentials. The appropriate code for an invalid input is 422 (Unprocessable Entity) or 400 (Bad Request).

Recommendation: Change the code passed to `messageCreator` for `handleIllegalRegexException` to 422 or 400. Also fix the `ExceptionType` — `ILLEGAL_REGEX_EXCEPTION` exists in the enum but `LOGIN_EXCEPTION` is used instead.

[MEDIUM] `handleAccessDeniedException` returns 401 instead of 403

Finding: `AccessDeniedException` signals that an authenticated user lacks permission. The correct HTTP status is 403 (Forbidden), not 401 (Unauthorized).

Recommendation: Change the status code for `handleAccessDeniedException` to 403. 401 should be reserved for unauthenticated requests.

[MEDIUM] `handleUserNotFoundException` returns a vague and incomplete message

Finding: The body is "Something went during the request." — a word is missing ('wrong'). More importantly, a 404 with a generic message is unhelpful to API consumers.

Recommendation: Fix the typo to 'Something went wrong during the request.' or provide a more specific message such as 'User not found.' for clarity.

[LOW] `USER_NOT_FOUND_EXPTION` enum value has a typo

Finding: In `ExceptionType.java` the value is `USER_NOT_FOUND_EXPTION` (missing 'E' in `EXCEPTION`). This is a silent bug — the description string 'User Not Found Exception' is correct, but the constant name is wrong.

Recommendation: Rename to `USER_NOT_FOUND_EXCEPTION`. Update all references (`GlobalExceptionHandler`).

[LOW] `deleteFile` throws `IllegiblePasswordException` — semantically wrong

Finding: `EFBoxFileController.deleteFile` declares throws `IllegiblePasswordException`. A file deletion operation should never throw a password-related exception. This looks like a copy-paste error.

Recommendation: Replace `IllegiblePasswordException` with `IllegalAccessException` or remove it entirely. The global handler already covers `AccessDeniedException` for auth checks.

[LOW] `ILLEGAL_ACCESS_EXCEPTION` still in enum but the exception was replaced

Finding: One of the commits changed `IllegalAccessException` to `AccessDeniedException` in the services and controllers, but `ILLEGAL_ACCESS_EXCEPTION` remains in the `ExceptionType` enum and is still referenced in `GlobalExceptionHandler.handleIllegalAccessException`. It's now dead code unless `IllegalAccessException` can still be thrown from somewhere.

Recommendation: Decide if `IllegalAccessException` (`java.lang`) can still appear in the codebase. If not, remove the handler and enum value to avoid confusion.

[LOW] `AccessDeniedException` import ambiguity

Finding: `EFBoxFileController` imports `org.springframework.security.access.AccessDeniedException` while `GlobalExceptionHandler` handles `java.nio.file.AccessDeniedException`. These are different classes. The Spring Security one is more appropriate here and will be intercepted by Spring Security's own filter chain before reaching the handler.

Recommendation: Standardise on `org.springframework.security.access.AccessDeniedException` throughout, or verify the handler catches the right type for each throw site.

[LOW] Zero test coverage on new code

Finding: SonarQube reports 0.0% coverage on new code. Exception handlers are straightforward to test with `MockMvc` — a test that triggers each exception type and asserts the HTTP status and body provides a safety net for future refactoring.

Recommendation: Add `@SpringBootTest` or `@WebMvcTest` tests that exercise at least the most critical handlers (`AccessDeniedException`, `UserNotFoundException`, `LoginException`).

Summary

Category	Count	Verdict
Strengths identified	8	
HIGH severity issues	0	
MEDIUM severity issues	4	
LOW severity issues	6	
SonarQube new issues (total)	11	9 are package naming (one fix covers all)
Test coverage on new code	0%	Recommended to add

Overall verdict: the architectural direction of this PR is solid and the code is significantly cleaner after the refactor. The main action items before merging are fixing the wrong `ExceptionType` values and the 401/403 status code confusion, correcting the typo in `USER_NOT_FOUND_EXPTION`, and resolving the `AccessDeniedException` import ambiguity. The package naming issue (EFbox vs efbox) is a pre-existing concern worth addressing separately but not a blocker for this PR.